1 The float ambiguity vehicle

The reference configurations (`config/golden/golden_field_refound.json` for real data, `config/golden/golden_sim_refound.json` for the simulated network) carry the carrier-phase ambiguities between epochs as float states of each station's local block. The invariant double-difference carry of the between-epoch carry section is off (`network.carry_invariant` false), and the loader refuses the float vehicle on a configuration that enables it. The invariant carry transports only the gauge-free chords and folds them through door 3 after the reduction (the three doors). The float vehicle transports every station's own ambiguity columns, conditional on the border, as rows of that station's block before the reduction. Gauge invariance then comes from each row's full border regression (Section 1.2) and from the held phase-datum basis (Section 1.5). The ambiguity process noise is zero: the parameter registry (`network/assembly/ParamRegistry`) takes it from `network.phase_leveling_q_amb` (class `leveling_ambiguity` of `network.process_noise`), loader default 0, kept by the reference configurations. The loader requires `network.phase_leveling` and `network.carry_conditional_coupling` (which itself requires `network.shared_carry_full_cov`) and forces the retired station-block switch `network.phase_leveling_local_block` off, because the vehicle supplies the joint block structurally.

1.1 The armed posture

Table 1 lists the settings both reference configurations arm that change the estimate or the emitted product and belong to the ambiguity carry (the `network.` prefix omitted).

Built and not armed. `network.amb_float_birth_var_m2` is absent, so its default 0 disables the birth floor of Equation (7); the drop-only floor `network.amb_float_drop_var_m2` is 0; `network.amb_float_declared_handover`, `network.carry_rebase_complete`, `network.carry_coupling_exact_cross`, `network.handover_stransform_before_rows`, `network.handover_stransform_dense`, `network.carry_coupling_clock_clone`, `network.carry_coupling_shared_history` and `network.ctu_switch_on_resume` are off. The clock clone and the shared history carry the satellite-clock uncertainty shared between the rows of one satellite; the self-contained clock level they were built for fails against an external clock product over several days (the two keys' rows in `docs/CONFIG_promotion_ledger.json`), so both stay off. The retired keys `network.phase_leveling_los_scale`, `network.carry_coupling_joint_whiten` and `network.amb_float_q_m2s` no longer exist.

1.2 The carried local column

The entry. `LocalCarry` (`network/driver/RunTypes`) is keyed by (station, satellite, signal) in the arc table (`net::ArcState`). After the accepted solve the harvest (`networkHarvestLocalCarries`, ambiguity rows in `networkHarvestAmbiguityStation` of `network/carry/LevelingCarry`) writes, for carried column c of station r (ambiguity in metres),

$$\hat{a}_c = x_c^{\mathrm{lin}} + \widehat{\delta x}_c, \qquad v_c = \left(\boldsymbol{N}_{rr}^{-1}\right)_{cc}, \qquad \boldsymbol{m}_c = \boldsymbol{N}_{sr}\,\boldsymbol{N}_{rr}^{-1}\boldsymbol{e}_c, \qquad C_{cc'} = \left(\boldsymbol{N}_{rr}^{-1}\right)_{cc'}: \tag{1}$$

`value_abs`, the pure conditional variance `var`, the border regression `coupling` (one term per border column j above `kCouplingSparsifyRelThresh` of the row maximum, with the solved value $\hat{s}_j$ and

setting = reference value (loader default)	mechanism	where
`ambiguity_vehicle` = `float_kf` (`level`)	joint local block; bounded drop	Sections 1.3 and 1.4
`phase_leveling` = true (false)	the station-local carry the vehicle folds	Section 1.2
`carry_conditional_coupling`, `carry_coupling_time_update`, `carry_coupling_gauge_project`, `carry_coupling_reinit_sources` = true (false)	border regression rows; scalar κ fallback; gauge restore; re-realised columns routed as departed	Sections 1.2 and 1.6
`shared_carry_full_cov` = true (false)	dense border carry $\boldsymbol{P}$	Section 1.6
`station_prior_before_pd` = true (false)	receiver DCB datum rows inside the tested block	Section 1.2
`phase_datum_prioritized`, `phase_datum_held_basis` = true (false)	per-(satellite, signal) anchor, elected and held	Section 1.5
`phase_datum_handover_continuous` = true (false); `phase_datum_handover_max_var_m2` = 0.01 (1.0)	successor pinned at its carried level	Section 1.5
`phase_anchor_carry_reref`, `phase_datum_held_cold_release`, `phase_datum_held_mark_cold` = true (false)	cold handover erased and marginalised, declared	Section 1.5
`phase_datum_gate_drop_erase` = true (false); `phase_datum_min_stations` = 2 (1)	erase at the coverage gate's drop edge	Section 1.4
`handover_stransform` = true (false)	clock-pivot S-transform of the border carry	Section 1.5
`rank_gate_tol` = 10^{-11} (10^{-9}); `sbasis_complete` = true (false)	gate and heal cutoff; completion of realised near-null directions	Section 1.5
`carry_coupling_exact_update`, `carry_rx_code_bias_in_block`, `amb_float_drop_decouple_always` = true (false)	exact time update; receiver code bias in the block; drop always decouples	Sections 1.4 and 1.6
`clock_temporal_anchor` = true (false); `clock_anchor_q` = 2.9×10^{-8} (10^{-4}); `clock_anchor_var_floor_m2` = 10^{-5} (0); `clock_anchor_guard_space` = `phase` (`clock`)	binding satellite-clock carry; guard on the phase-visible clock	Section 1.6
Event-response and emitted-product settings, documented in the event-response and emitted-product section:		
`cycle_slip.relevel_free_epoch`; `cycle_slip.mw_reflag_skip`; `cycle_slip.windup_restart_refloat`; `estimation.accumulate_phase_windup`	free-epoch latch; one slip frees an arc once; wind-up restart response	
`readmit_seed_horizon_s` = 3900 (0); `readmit_seed_mirrors`; `cycle_slip.gap_reentry_relevel`; `cycle_slip.gap_reentry_per_signal`; `cycle_slip.tracker_modeled_first`	re-admission seeds; per-row and per-signal gap re-entry	
`gross_los_screen`; `gross_los_cold_bound_m` = 12 (0)	acting code-row tier per line of sight	
`emit_pb_hold`; `emit_pb_hold_phase_clock`; `closed_loop.pb_hold_presence_flip`	generation hold; level step moved to the held bias; presence edges at the user	
`clock_disc_jump_test`; `emit_clock_frame_hold`; `emit_clock_disc`; `emit_ura`	clock step test; held clock frame; clock discontinuity counter; quality channel	

Table 1: Settings of the float ambiguity vehicle armed by both reference configurations. Values in parentheses are loader defaults.

the datum-frame variance p_j, 0 when pinned) and the cross-covariance block `blk`, all from one local solve $\boldsymbol{N}_{rr}^{-1}\boldsymbol{e}_c$ (`stationRegression`). The entry also keeps `var_first` (the first-stash v_c), `var_birth`, the birth stamps and `relevel_s`, the time of the last re-level. The anchored column of each family is excluded at inject and harvest: its value is the datum.

Why the regression row. With $\boldsymbol{u}_r = \boldsymbol{x}_r + (\boldsymbol{N}_{rr}^r)^{-1}\boldsymbol{N}_{sr}^{r\mathrm{T}}\boldsymbol{x}_s$, completing the square in each station's block of the arrowhead system splits the posterior quadratic form exactly:

$$(\boldsymbol{x}-\hat{\boldsymbol{x}})^{\mathrm{T}}\boldsymbol{N}(\boldsymbol{x}-\hat{\boldsymbol{x}}) = (\boldsymbol{x}_s-\hat{\boldsymbol{x}}_s)^{\mathrm{T}}\tilde{\boldsymbol{N}}_{ss}(\boldsymbol{x}_s-\hat{\boldsymbol{x}}_s) + \sum_r (\boldsymbol{u}_r-\hat{\boldsymbol{u}}_r)^{\mathrm{T}}\boldsymbol{N}_{rr}^r(\boldsymbol{u}_r-\hat{\boldsymbol{u}}_r). \tag{2}$$

Door 1 carries the border term. The vehicle carries the second: component c of $\boldsymbol{u}_r$ is $x_c + \boldsymbol{m}_c^{\mathrm{T}}\boldsymbol{x}_s$, with conditional covariance $(\boldsymbol{N}_{rr}^r)^{-1}$, and becomes the pseudo-observation

$$\delta x_c + \boldsymbol{m}_c^{\mathrm{T}}\boldsymbol{x}_s = \left(\hat{a}_c - x_c^{\mathrm{lin}}\right) + \boldsymbol{m}_c^{\mathrm{T}}\hat{\boldsymbol{s}} + \epsilon_c, \qquad \epsilon_c \sim N(0,\sigma_c^2), \tag{3}$$

with $\sigma_c^2 = v_c$ before the time update (Section 1.6). A row stating $\hat{a}_c$ at v_c with no border coefficients would state the border along $\boldsymbol{m}_c$ a second time, on top of door 1.

Linearisation invariance. Replacing x_c^{lin} by $x_c^{\mathrm{lin}} + \delta$ maps $\delta x_c \mapsto \delta x_c - \delta$ and the right-hand side of Equation (3) by $-\delta$. Storing the absolute $\hat{a}_c$ makes a shift of the linearisation point an exact solve no-op.

Gauge invariance. Every data row of station r satisfies $\boldsymbol{A}_{s,r}\boldsymbol{g}_s + \boldsymbol{A}_{r,r}\boldsymbol{g}_r = \boldsymbol{0}$ along each structural gauge direction, hence $\boldsymbol{N}_{sr}^{r\mathrm{T}}\boldsymbol{g}_s + \boldsymbol{N}_{rr}^r\boldsymbol{g}_r = \boldsymbol{0}$ and

$$\boldsymbol{m}_c^{\mathrm{T}}\boldsymbol{g}_s = -\,g_r(c)\,: \tag{4}$$

the row of Equation (3) is blind to every gauge, provided the datum-pinned terms are kept. Without them every row observes the clock gauge and the null-space gate withholds.

Which door. The row's increment lands in station r's own $\boldsymbol{N}_{ss}^r$, $\boldsymbol{N}_{sr}^r$, $\boldsymbol{N}_{rr}^r$, so the arrowhead is preserved. It is an observation of that station: `appendCouplingRows` (`network/carry/Coupling`) appends it to $\boldsymbol{A}_{s,r}$, $\boldsymbol{A}_{r,r}$, $\boldsymbol{W}_r$, $\boldsymbol{y}_r$, and it enters through `addStation`. Like door 2 it precedes the reduction; unlike door 2 it also adds to the station's border blocks. The phase-datum pins are $\boldsymbol{A}_r$ rows of weight 10^{10} with right-hand side $t - x_c^{\mathrm{lin}}$. No carried prior folds before `network.phase_leveling_warmup_epochs` emitted epochs (default 6). The rows are built in the parallel station produce of PASS B as the steps `kStationBuildSteps` and `kStationCarrySteps` of `network/assembly/StationBuild`.

Station prior before the PD test. `addStation` rejects a station whose $\boldsymbol{N}_{rr}$ factor fails the positive-definiteness pivot test. With `network.station_prior_before_pd` the receiver DCB datum rows ($\Lambda = 1/\sigma^2$, $\sigma =$ `network.rx_code_bias_datum_sigma_m`, $\eta = -\Lambda\, x_c^{\mathrm{lin}}$) are folded into a copy of $\boldsymbol{N}_{rr}$, $\boldsymbol{u}_r$ before the test (`StationPrior`) and withheld from the later `addStationPrior` call; without the setting, stations were rejected for that missing row. On the reference configurations every carried ambiguity row and in-block receiver code-bias row is also a row of $\boldsymbol{A}_{r,r}$, so the test judges the block the solve uses.

1.3 The joint local block

Under `float_kf`, `NetworkKnobs::pl_local_block` follows the vehicle and `CouplingRowBuilder::buildLocalBlock` folds each station's carried columns jointly (given `carry_conditional_coupling`). For the columns C that pass the inject predicates (present, not gauge-pinned, not pending a reseed, finite x_c^{lin}, a live entry), with the row ingredients $(\boldsymbol{m}_c, y_c, \sigma_c^2)$ of Section 1.6,

$$(\boldsymbol{\Sigma}_C)_{ii} = \sigma_{c_i}^2, \qquad (\boldsymbol{\Sigma}_C)_{ij} = \begin{cases} C_{c_i c_j} & \text{same harvest block,} \\ 0 & \text{otherwise (counted),} \end{cases} \qquad \boldsymbol{\Sigma}_C = \boldsymbol{L}\boldsymbol{L}^{\mathrm{T}}, \tag{5}$$

and the station receives the unit-weight rows $\boldsymbol{L}^{-1}[\,\boldsymbol{I}_C \mid \boldsymbol{M}_C \mid \boldsymbol{y}_C\,]$ (`whitenLocalBlock`). Each block of $\boldsymbol{\Sigma}_C$ is a principal submatrix of one posterior, so $\boldsymbol{\Sigma}_C$ is positive-definite by construction; after one ridge retry a failed factor reverts the station to per-column rows, counted. A column refloated within the epoch has its row and column deleted before the factor: an exact marginalisation. The receiver code bias joins the same block (Section 1.6, paragraph on the receiver code bias in the block).

$\boldsymbol{\Sigma}_C$ is the joint conditional covariance of $\boldsymbol{u}_C$ with every non-carried local integrated out. Independent rows fold $\mathrm{diag}(\boldsymbol{\Sigma}_C)^{-1}$. For n columns of variance σ^2 and pairwise correlation ρ, along $\boldsymbol{v} = \mathbf{1}/\sqrt{n}$,

$$\boldsymbol{v}^{\mathrm{T}}\boldsymbol{\Sigma}_C^{-1}\boldsymbol{v} = \frac{1}{\sigma^2\,[1 + (n-1)\rho]}, \qquad \boldsymbol{v}^{\mathrm{T}}\mathrm{diag}(\boldsymbol{\Sigma}_C)^{-1}\boldsymbol{v} = \frac{1}{\sigma^2}, \tag{6}$$

an over-count of $1 + (n-1)\rho$ along the directions the non-carried locals share: the slant ionosphere of a line of sight's two bands, and the receiver clock, zenith wet delay and receiver DCB of every column. The block is exact in every local-local direction given the border.

1.4 Birth prior and the bounded drop

Birth. A new arc folds no carried prior in its first epoch; a prior from the same epoch's data would count it twice. The first harvest records $\sigma^2_{\mathrm{first}} = v_c$ and sets, once,

$$\sigma^2_{\mathrm{birth}} = \max\{\sigma^2_{\mathrm{first}},\ f(e)\}, \qquad f(e) = \frac{\sigma^2_{\mathrm{zen}}}{\left[\sin \max(e,\, 5^\circ)\right]^{2p}} \tag{7}$$

(`ambFloatBirthFloor`), with σ^2_{zen} = `network.amb_float_birth_var_m2` and p = `filter.code_elevation_power`, the code weights' mapping, because the floor states what a code-determined level is worth. An unknown elevation reads as zenith; $\sigma^2_{\mathrm{zen}} \le 0$ gives $f \equiv 0$. On the reference configurations $\sigma^2_{\mathrm{birth}} = \sigma^2_{\mathrm{first}}$.

The drop. A slip lowers confidence in the level and keeps the arc's identity. `ambFloatDrop` (through `ambFloatDropFloored`) sets

$$\sigma^2_{\mathrm{tgt}} = \max\{\sigma^2_{\mathrm{birth}},\ \sigma^2_{\mathrm{first}},\ f_b(e),\ f_d(e)\}, \qquad \sigma_c^2 < \sigma^2_{\mathrm{tgt}} \ \Rightarrow\ \sigma_c^2 \leftarrow \sigma^2_{\mathrm{tgt}},\ \ \boldsymbol{m}_c \leftarrow \varnothing,\ \ C_{c\cdot} \leftarrow \varnothing, \tag{8}$$

with f_b, f_d the floor at the birth floor and at `network.amb_float_drop_var_m2`. The age clock restarts; $\hat{a}_c$, `var_first`, `var_birth`, the birth stamps, `last_seen_s` and `n_folds` are kept. A drop only loosens: for $\sigma_c^2 \ge \sigma^2_{\mathrm{tgt}}$ it is idempotent, and `network.amb_float_drop_decouple_always` makes that branch clear the regression and block row too. Without it an arc slipping one epoch after birth kept its pre-slip regression and cross terms, and under the exact update its stale value went into the satellite clocks. On the reference configurations $\sigma^2_{\mathrm{tgt}} = \sigma^2_{\mathrm{first}}$; the prior still folds in the same epoch.

Causes. `networkAmbFloatApplyDrops` (`networkAmbFloatDropEpoch` in the epoch spine) runs between the PASS-A slip scans and PASS B, so the inject folds at birth strength where the leveling vehicle skips: `relevel_free` (the latched free set of `network.cycle_slip.relevel_free_epoch`: slip detectors, gap re-entry and wind-up restart) and `stale_gap` (gap above `network.phase_leveling_max_gap_s`, reference value 900 s, default 120 s). `networkPostFitRaim` drops on `t4_attrib` and `t3_consensus`, where the leveling vehicle erases. Under the held basis each latched arc is stamped in the station row's `amb_flag_ep`. The causes are the constants of `AmbDropCause`. An optional maturity gate on the isolated-arc refloats of post-fit RAIM (`network.raim_refloat_maturity_gate`, off) holds a refloat on an arc younger than its horizon; the gate and its telemetry belong to the quality-control chapter (the network RAIM section), and the vehicle sees it only as the absence of that drop.

The erases that remain. The post-harvest prune erases an arc unseen past `network.phase_leveling_max_gap_s`. On the reference configurations three more responses erase carried ambiguities: the gross line-of-sight screen (`network.gross_los_bound` off; the gross line-of-sight section), the anchor re-reference (Section 1.5), and the coverage gate's drop edge, where the survivor's carried ambiguity would absorb the removed bias.

Both edges. A mechanism that changes which information reaches which states is an event at both edges. The decoupling edge is bounded: a free epoch opens a direction the border re-balances into, mainly the satellite clock, since the anchor pin holds the phase bias; the drop removes information down to a stated level. The re-coupling edge re-validates: the arc re-enters as an uncoupled row at σ^2_{tgt}, its cross terms read as pair misses until the next harvest, and it re-earns its variance from the data. Drops are counted by cause and traced in the trace file `network_amb_state.csv`.

1.5 The held S-basis

The family gauge. Each (satellite i, signal s) family carries the gauge

$$G(c): \qquad a^i_{r,s} \mapsto a^i_{r,s} - c \;\; \forall r, \qquad b^i_s \mapsto b^i_s + c, \tag{9}$$

with b^i_s the satellite phase bias. The prioritized datum (`networkElectPhaseAnchors` in `network/solve/PhaseDatum`, called by `networkPhaseAnchorEpoch`) pins $a^i_{r^\star,s} = t$ at one anchor $r^\star$. Two realisations differ by an exact S-transform [1],

$$\boldsymbol{x}(r', t') = G(c)\,\boldsymbol{x}(r^\star, t), \qquad c = \hat{a}^i_{r',s}(r^\star, t) - t', \tag{10}$$

and every carried reference of the family (each $\hat{a}$, the carried phase bias, each $\hat{s}_j$ on its column) lives in the realisation it was harvested in.

Held basis. Under `network.phase_datum_held_basis` a retained incumbent re-pins at the value frozen at its election (`PhaseBasisEntry::value`): $c = 0$ within a tenure. A datum re-derived after a drop or a flag would apply $G(c)$, $c \neq 0$, a phase-bias step the user re-converges through. The basis ambiguity is a held constant outside the carry; every other ambiguity is a live state. A screen flag on the incumbent's own column within `network.phase_anchor_flag_quiet_ep` epochs (default 5) deposes it at the election. A new election ranks by elevation the candidates aged at least `network.phase_anchor_min_age_ep` epochs (default 10), unflagged and with a usable ledger entry; failing that, by elevation alone (a cold pin), then coverage. A slip response that frees a family's anchor arc bans that incumbent for the epoch (`network.cycle_slip.anchor_slip_handover`), so the election hands the family to a successor pinned at its own carried level.

Handover. `AnchorHandoverLedger` holds each station's previous-epoch $\hat{a}$ (written ℓ) and conditional variance; an entry is usable when $0 < \text{var} \le$ `network.phase_datum_handover_max_var_m2`. A continuous handover pins at $t' = \ell_{r'}$: c is the successor's one-epoch revision, bounded by the ceiling, and the family does not step. A cold handover pins at $t' = 0$: $c = \ell_{r'} =: \Delta$. The reference configurations answer a cold handover, a first sighting or a bridge exit by erasing the family's carried ambiguities and marginalising its phase-bias key out of the border carry for one epoch (`network.phase_anchor_carry_reref`, `network.phase_datum_held_cold_release`); with `network.phase_datum_held_mark_cold` the emitted product declares a handover if and only if it is cold. The exact alternative (`declaredHandoverShiftCarry`, behind `network.amb_float_declared_handover`, off) applies $G(\Delta)$ to the carry,

$$\hat{a}^i_{r,s} \leftarrow \hat{a}^i_{r,s} - \Delta, \qquad \hat{b}^i_s \leftarrow \hat{b}^i_s + \Delta, \qquad \hat{s}_j \leftarrow \hat{s}_j + \Delta \;\; (j = b^i_s), \tag{11}$$

variances untouched; moving one reference alone injects the frame step as row bias. The setting `network.carry_rebase_complete` (off) extends the shift to the sites the mechanism table lists as owed.

Clock pivot. At a clock-pivot swap $p \to q$, `network.handover_stransform` (`network/solve/HandoverStransform`) re-expresses the border carry along the structural clock null direction $\boldsymbol{v}_c$,

$$c_0 = x_q / v_c[q], \qquad x_i \leftarrow x_i - v_c[i]\, c_0, \qquad \text{var}_i \leftarrow \text{var}_i + v_c[i]^2 \,\text{var}_q, \tag{12}$$

cross term dropped, and releases that constellation's one-epoch carry skip (it needs `network.scoped_handover_response`, armed).

Rank gate and heal. `NetworkSbasis::assertNullSpaceRank` requires

$$n_{\text{null}} = \#\big\{k : \lambda_k(\tilde{\boldsymbol{N}}_{ss}) < \tau \, \max(\lambda_{\max}, 1)\big\}, \qquad \tau = \texttt{network.rank_gate_tol}, \tag{13}$$

to equal the declared S-basis dimension (solve stage 8, `networkRankGate`). On failure, `healNearNullPhaseBias` soft-anchors toward 0 at $\sigma = 1$ m every phase-bias column with absolute component above 0.3 in a direction below the same cutoff, through `addSharedPrior` (door 1, no second reduction), and re-gates once; a second failure withholds. A held basis whose anchor station is rejected leaves its column near-null, and the heal makes that a bounded one-epoch degradation. Gate, heal and estimability dump read one τ. With the exact update the level keeps a code-scale variance in the first minutes, and at $\tau = 10^{-9}$ weak but real phase-bias common modes were counted as null; the reference configurations set 10^{-11} and also arm `network.sbasis_complete`, which constrains a realised near-null direction beyond the declared dimension along its own eigenvector (solve stage 7) instead of withholding.

1.6 The coupling time update

A row harvested at epoch $k-1$ states $l = r + \boldsymbol{m}^{\mathrm{T}}\boldsymbol{s} + e$, $e \sim N(0, v)$, given that epoch's border $\boldsymbol{s}$. The border moves, $\boldsymbol{s}^+ = \boldsymbol{s} + \boldsymbol{w}$, $\boldsymbol{w} \sim N(\boldsymbol{0}, \boldsymbol{Q})$, and the border fold re-injects it at $N(\hat{\boldsymbol{s}}, \boldsymbol{P} + \boldsymbol{Q})$, with $\boldsymbol{P}$ the stashed dense posterior and $\boldsymbol{Q}$ the fold's own inflation. The row must be re-expressed on $\boldsymbol{s}^+$ with the same $\boldsymbol{Q}$.

The scalar update. `scalarRowIngredients` shrinks each present term with $q_j\Delta t > 0$, $p_j > 0$:

$$\kappa_j = \frac{p_j}{p_j + q_j\Delta t}, \qquad m_j \leftarrow \kappa_j m_j, \qquad \sigma^2 \mathrel{+}= m_j^2\,\kappa_j\,q_j\Delta t; \tag{14}$$

a departed or re-realised term adds $m_j^2 p_j$ and is deleted, and `tuGaugeRestore` spreads each (constellation, kind) deficit over the family's present terms so Equation (4) holds. q_j is the registry's border q (`couplingQ`); for a clock, `filter.process_noise.sat_clock_q`, reference value $1 \times 10^{-3}\,\mathrm{m^2\,s^{-1}}$. The fold anchors the same clock at `network.clock_anchor_q` with floor `network.clock_anchor_var_floor_m2` (reference value $1 \times 10^{-5}\,\mathrm{m^2}$). At the floor and $\Delta t = 30$ s, $\kappa_{\mathrm{clk}} \approx 3 \times 10^{-4}$ while $\kappa_{\mathrm{pb}} \approx 1$: the row loses its own clock coefficient.

The level lock. In the single-station, two-band model (phase $\phi_j = \rho + T + dt_r - c + p_j + N_j - \alpha_j I$, code $P_j = \rho + T + dt_r - c + b_j + \alpha_j I$, $\alpha_1 = 1$), the code rows give $\partial I/\partial c = \kappa' = \sum_j w_j\alpha_j / \sum_j w_j\alpha_j^2$, and

$$\text{exact row: } N_j - (1 + \alpha_j\kappa')\,c + p_j = r_j, \qquad \text{shrunk row: } N_j + p_j = r_j. \tag{15}$$

With the phase row, the exact row constrains $\alpha_j(I - \kappa' c)$ and leaves c to the code. The shrunk row gives $c + \alpha_j I = r_j - \phi_j + \rho + T + dt_r$ for $j = 1, 2$, which fixes the satellite clock level at phase precision at every station and epoch. The level is estimable from code only; stated at phase precision, it freezes (`docs/refound/CTU_coupling_time_update_2026-09-28.md`).

The exact update. `network.carry_coupling_exact_update` (`network/carry/CouplingExact`) splits a row's stashed terms into S, the columns the fold carries with a finite inflation $\boldsymbol{Q}_S$, and D, the rest (departed, re-initialised, re-referenced, skipped). Over $K = S \cup D$,

$$\begin{aligned} \boldsymbol{m}_{\mathrm{new}} &= (\boldsymbol{P}_{SS} + \boldsymbol{Q}_S)^{-1}\boldsymbol{P}_{S,K}\,\boldsymbol{m}_K, \qquad \operatorname{Var} e' = v + \boldsymbol{m}_K^{\mathrm{T}}\boldsymbol{B}\,\boldsymbol{m}_K, \\ \boldsymbol{B} &= \boldsymbol{P}_{KK} - \boldsymbol{P}_{K,S}(\boldsymbol{P}_{SS} + \boldsymbol{Q}_S)^{-1}\boldsymbol{P}_{S,K}. \end{aligned} \tag{16}$$

With $\boldsymbol{F}_S = (\boldsymbol{P}_{SS} + \boldsymbol{Q}_S)^{-1}$, $\boldsymbol{g} = \boldsymbol{Q}_S\boldsymbol{m}_S$, $\boldsymbol{u} = \boldsymbol{P}_{S,D}\boldsymbol{m}_D$, $\boldsymbol{z} = \boldsymbol{F}_S\boldsymbol{u}$ and $\boldsymbol{F}_S\boldsymbol{P}_{SS} = \boldsymbol{I} - \boldsymbol{F}_S\boldsymbol{Q}_S$, `couplingExactRow` evaluates both halves free of cancellation:

$$\boldsymbol{m}_{\mathrm{new}} = \boldsymbol{m}_S - \boldsymbol{F}_S\boldsymbol{g} + \boldsymbol{z}, \qquad \boldsymbol{m}_K^{\mathrm{T}}\boldsymbol{B}\,\boldsymbol{m}_K = \boldsymbol{m}_{\mathrm{new}}^{\mathrm{T}}\boldsymbol{g} + \boldsymbol{g}^{\mathrm{T}}\boldsymbol{z} + \left(\boldsymbol{m}_D^{\mathrm{T}}\boldsymbol{P}_{DD}\boldsymbol{m}_D - \boldsymbol{u}^{\mathrm{T}}\boldsymbol{z}\right), \tag{17}$$

the variance clamped at 0 (counted). With D empty, $\boldsymbol{Q}_S = \boldsymbol{0}$ returns the harvested row and a diagonal $\boldsymbol{P}$ returns Equation (14). The difference is in D: a decoupled or departed clock correlated with its phase biases along the level gives $m_{\mathrm{new},p} \approx m_p + (P_{pc}/P_{pp})\,m_c$, so the row keeps its co-movement with the level.

The assembled row. `exactRowIngredients` emits $\boldsymbol{m}_{\mathrm{new}}$ on S with right-hand-side terms $m_{\mathrm{new},j}\hat{s}_j$, $\hat{s}_j$ the mean the fold centres its prior on. A datum-pinned term is kept as harvested; a present term with no dense row takes Equation (14) at the fold's inflation; a departed one adds $m_j^2 p_j$; SH-VTEC terms keep the scalar update. Per constellation the clock coefficients the update removes (harvested minus emitted) go to the stash's pinned pivot clock, so the row keeps its identity along the declared clock gauge. The variance

$$\sigma_c^2 = v_c + \textstyle\sum_{\mathrm{dep}} m_j^2 p_j + \sum_{\mathrm{scalar}} m_j^2\kappa_j q_j\Delta t + \boldsymbol{m}_K^{\mathrm{T}}\boldsymbol{B}\,\boldsymbol{m}_K \tag{18}$$

is the diagonal of $\boldsymbol{\Sigma}_C$ in Equation (5). Only rows harvested at the stash of $\boldsymbol{P}$ take this route; others, and epochs whose $\boldsymbol{P}_{SS} + \boldsymbol{Q}_S$ does not factor, take the scalar update, counted. $\boldsymbol{Q}_S$ is the fold's own

decision, made once before PASS B (`networkCarryDecide` in `network/state/CarryDisposition`, `networkCouplingExactPrepare` in `network/state/BorderPrior`: $q = \texttt{var_eff} - P_{pp}$), so rows and fold read one $\boldsymbol{Q}$. The covariance $\boldsymbol{m}_i^{\mathrm{T}} \boldsymbol{B}\, \boldsymbol{m}_k$ between rows is dropped; the within-station half (`network.carry_coupling_exact_cross`) is built and off.

Information loss. Under the scalar update with the loose clock rate, a young arc's ambiguity information is discarded at every inject; under the exact update the steady loss is the anchored clock's walk. The per-arc accounting of the variance budget is an open item (`docs/refound/AIL_ambiguity_information_loss_2026-10-01.md`).

Receiver code bias in the block. The diagonal receiver code-bias carry folded each column at its conditional variance with no border coefficients, the double count that Equation (3) avoids; the border information along each satellite's level roughly doubled every epoch. `network.carry_rx_code_bias_in_block` harvests these columns with their regression and cross covariances into the block of Equation (5), with no diagonal fallback.

Temporal clock anchor. The guarded anchor, `network.clock_temporal_anchor` (`network/solve/ClockAnchor`), folds each satellite clock of the border carry at

$$v_{\mathrm{eff}} = \begin{cases} \max\{v + q_a\,\Delta t_i,\ v_{\mathrm{floor}}\} & \text{anchored,} \\ v + q_c\,\Delta t & \text{loose,} \end{cases} \qquad \Delta t_i = t - t_{\mathrm{last}}(i), \tag{19}$$

$q_a =$ `network.clock_anchor_q`, $v_{\mathrm{floor}} =$ `network.clock_anchor_var_floor_m2`, $q_c =$ `sat_clock_q`; an anchored carry older than `network.clock_anchor_max_gap_s` (default 120 s) decouples. Without the anchor, at the reference q_c the carried prior spread $\sqrt{q_c \Delta t}$ exceeds the data's own per-epoch spread, the prior does not bind, and the emitted clock is re-drawn from each epoch's data as a random walk. Once the anchor binds, the emitted walk is $\sqrt{q_a \Delta t}$, because for the scalar carry $\mathrm{Var}(\hat{x} - \mu) = P^- - P^+$ and $\mathrm{Var}(\hat{x}_k - \hat{x}_{k-1}) \to q_a \Delta t$. The guard standardises the common-removed residual by $\sqrt{\max(P^- - P^+, \sigma_f^2)}$, $\sigma_f =$ `network.clock_anchor_sigma_floor_m`. A trip at `network.clock_anchor_k` (default 6), or above `network.clock_anchor_abs_m` (default 2 m), sets the satellite loose, withdraws its prior, re-solves before emission and marks the emitted product; the re-solve needs a carry redundancy of at least 0.05. The guard runs as solve stage 12, between the primary solve and the within-epoch RAIM tier, so that a prior fault is refloated before the tier can attribute it to data. Re-coupling needs `network.clock_anchor_recouple_epochs` (default 4) in-band epochs. Both edges are events: the decoupling reasons are counted separately (statistic, backstop, stale carry, datum change, refloat by another tier), and the re-coupling edge is a counted event that the emitted product does not mark, because re-engaging a prior centred on the last posterior steps no emitted value.

Guard space. With the exact update the level carries a code-scale variance while the phase-visible combinations are phase-precise, and a clock-space statistic trips on legitimate level convergence. Under `network.clock_anchor_guard_space` = `phase` the trip reads

$$d = c - \textstyle\sum_j w_j\, p_j, \qquad \boldsymbol{w} = \arg\min \|\boldsymbol{w}\| \text{ s.t. } \sum_j w_j = 1,\ \sum_j w_j \lambda_j^2 = 0, \tag{20}$$

one phase bias per band, constellation median removed, $\sqrt{S}$ floored at `network.clock_anchor_phase_sigma_floor_m` (default 0.02 m). The level (a common shift of c and every p_j) and the

ionosphere direction (p_j by $\alpha_j\delta$) leave d unchanged; GPS L1/L2 gives $\boldsymbol{w} = (2.5457, -1.5457)$. A satellite without a two-band phase statistic in an epoch is tested on the backstop alone. The backstop stays on the clock.

Neighbouring clock mechanisms. Four mechanisms act on the same clock columns and are described where their mathematics lives. The cold receiver-clock pre-seed (`network.cold_clock_seed`, armed) removes the free-running receiver-clock offset from cold ambiguity seeds (the state-initialisation section). The clock runaway check (`network.clock_runaway_check`, armed) drops from the emission a constellation whose median clock differs from the global median by more than `network.clock_runaway_mag_m` (1000 m); its drift ceiling needs the two-state clock model, which is off (the emission-gate section). The clock step test, the held clock frame and the clock discontinuity counter act on the emitted clocks (the emitted-clock section). The cold-start a-priori clock anchor (solve stage 10, `networkFoldColdClockAnchor`) pulls each clock toward the a-priori product clock with $\sigma =$ `network.apriori_clock_anchor_sigma_m`, or `filter.initial_std.sat_clock` when unset; it binds only below 100 m, and the reference value of 1000 m leaves it off.

Fresh convergence only. A snapshot written without the exact update, the in-block receiver code bias, the drop decouple or the phase-space guard is not resumed into a run that arms them unless `network.ctu_switch_on_resume` (default off). That seam failed its gate: the restored ambiguities, and the slant ionosphere they tie, had absorbed the frozen level error and hold it; the ambiguity-side response is not built (`docs/refound/CTU2_level_recursion_followup_2026-09-29.md`).

1.7 Status

The float vehicle with the held basis, the station prior before the PD test, the exact update with its companions and the level-aware hold are armed on both reference configurations. Their record is in `docs/refound/RF-1_decision_2026-09-14.md`, `RF-2_held_basis_2026-09-15.md`, `GRB_golden_rebase_2026-09-22.md`, `CTU_coupling_time_update_2026-09-28.md` and `EXL_exact_update_lah_promotion_2026-10-01.md`. The registry and the per-epoch fold decision are consolidations that change no output (`PRM_parameter_registry_2026-09-29.md`).

The baseline after the latest arming (`docs/refound/BOF_built_off_follow_through_2026-10-28.md`), over a 6 h run on the reference configurations: the test station WTZZ00DEU on real data, which the network does not use, has a median 3-D position error per hour of 0.059, 0.012, 0.023, 0.012, 0.017 and 0.012 m; on the simulated network the test station SZZ000SIM, graded from the run start, has 0.137, 0.072, 0.031, 0.024, 0.026 and 0.018 m. The first hour is the convergence hour and the settled claims come from hours 2 to 6.

Open: the birth floor, set by no reference configuration; the switch-on seam's ambiguity-side response and the guard's answer to a phase-bias event; the information budget; the residual non-finite-input station rejections; and the station level under three frequency bands, for which the anchor's level directions held in the phase datum are derived offline and not built (`docs/refound/R1D_station_level_datum_2026-10-23.md`, `ALD_anchor_level_datum_2026-10-23.md`). The register of open items is `docs/status/OPEN_ITEMS.md`.

References

[1] P. J. G. Teunissen. "Generalized inverses, adjustment, the datum problem and S-transformations". In: *Optimization and Design of Geodetic Networks* (1985), pp. 11–55.